Vulnerability Assessment & Penetration Testing (VAPT)

WEEK 1 – Lab Setup & Basic Scanning Report

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Environment: Host-Only / Internal Network (Isolated Virtual Lab)

Tools Used: Kali Linux, Nmap, OpenVAS, Docker

1. Objective

The objective of Week 1 was to establish a secure and isolated penetration testing lab environment and perform initial reconnaissance on intentionally vulnerable systems to identify open ports, services, and potential security weaknesses.

2. Virtual Lab Setup & Network Configuration

A virtual lab was configured using VMware Workstation / VirtualBox. The network adapter was set to Host-Only or Internal Network mode to ensure full isolation from external or production environments.

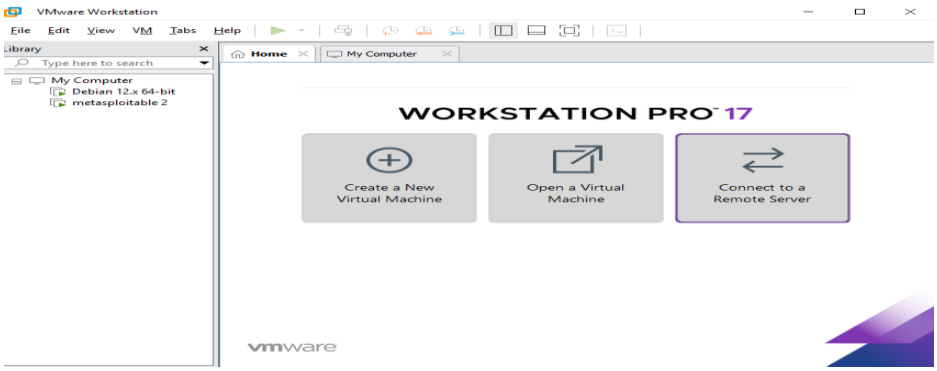
Configuration Details:

• Hypervisor: VMware / VirtualBox

• Network Mode: Host-Only / Internal Network

• External Connectivity: Disabled

• Lab Isolation: Confirmed



3. Attacker Machine – Kali Linux Setup

Kali Linux was deployed as the attacker machine. After installation, the system was updated to ensure the latest packages and security tools were installed.

System Update Command:

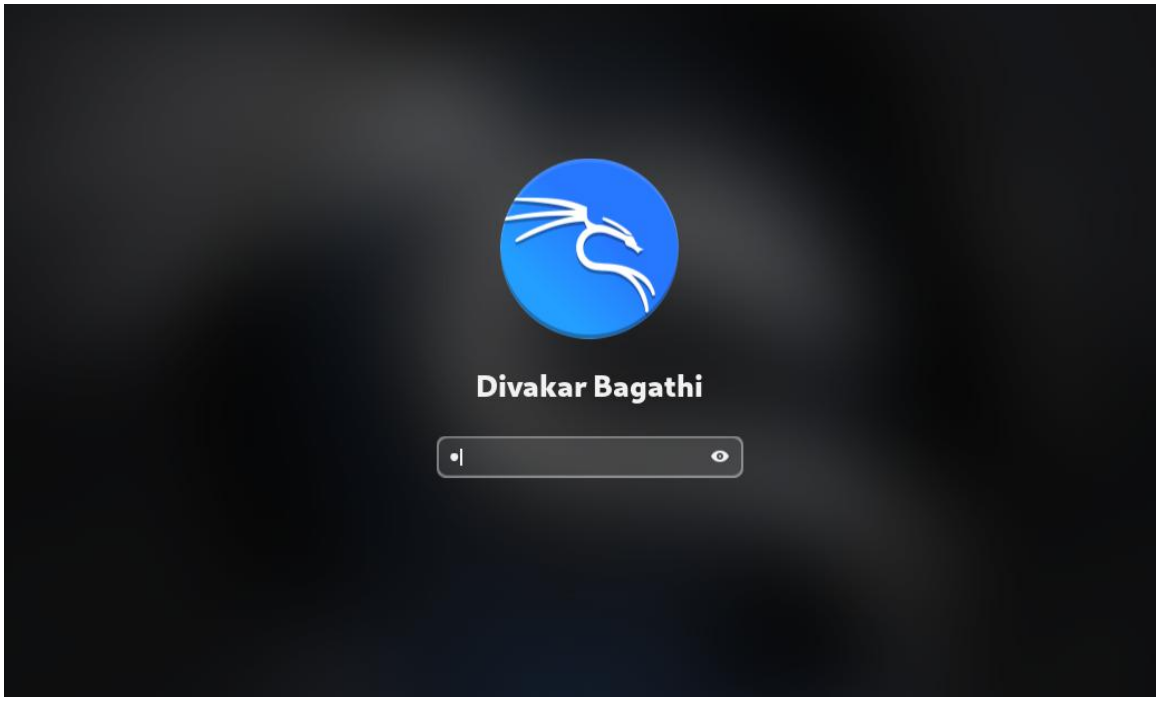
sudo apt update && sudo apt upgrade -y

Verified Tools:

• Nmap – Network scanning and service enumeration

• OpenVAS – Vulnerability scanning

• Docker – Web application deployment



4. Target Machine Deployment – Metasploitable 2

Metasploitable 2 was installed as an intentionally vulnerable Linux system to simulate real-world security weaknesses.

Default Credentials:

• Username: msfadmin

• Password: msfadmin

IP Address Identified: 192.168.230.131

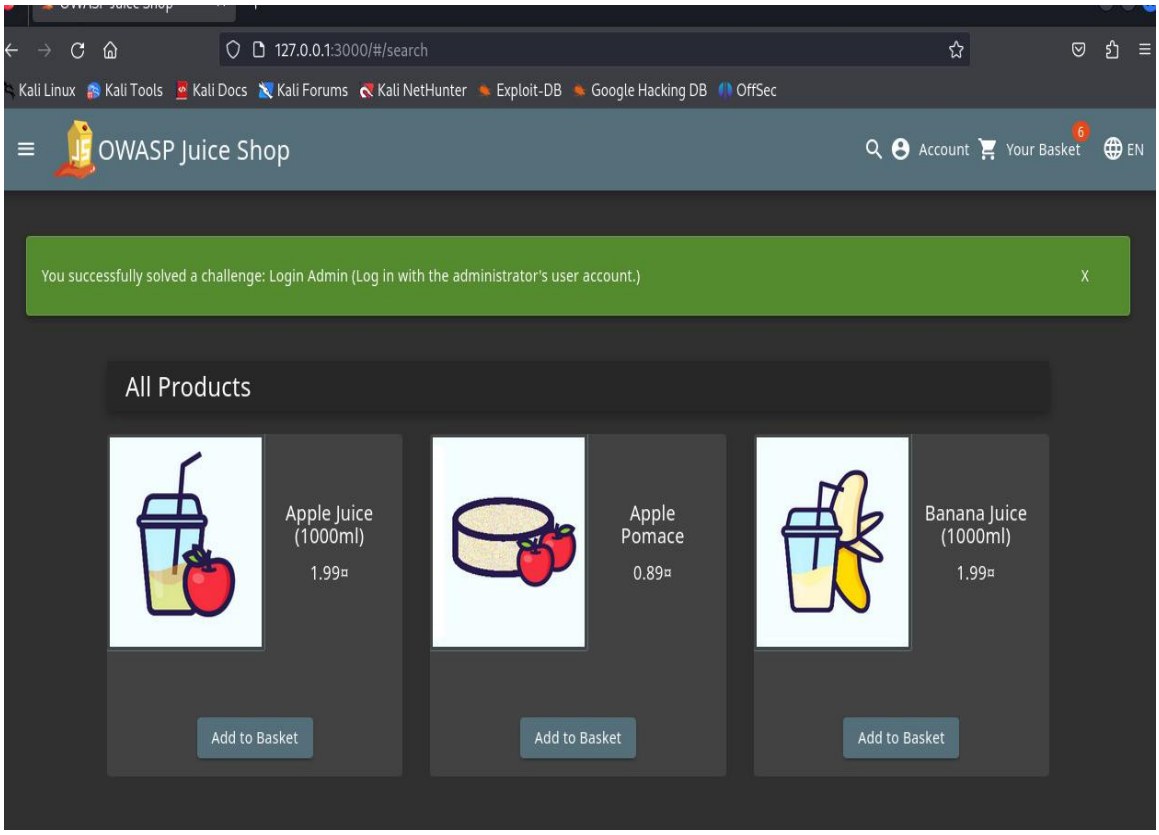
```
 * Starting deferred execution scheduler atd                              [ OK ]
 * Starting periodic command scheduler crond                              [ OK ]
 * Starting Tomcat servlet engine tomcat5.5                               [ OK ]
 * Starting web server apache2                                            [ OK ]
 * Running local boot scripts (/etc/rc.local)
nohup: appending output to `nohup.out'
nohup: appending output to `nohup.out'
                                                                          [ OK ]

Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started


metasploitable login: _
```

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:ad:2e:80
          inet addr:192.168.230.131  Bcast:192.168.230.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fead:2e80/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:50 errors:0 dropped:0 overruns:0 frame:0
          TX packets:64 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4999 (4.8 KB)  TX bytes:7008 (6.8 KB)
          Base address:0x2000 Memory:fd5c0000-fd5e0000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$
```

5. OWASP Juice Shop Deployment

OWASP Juice Shop was deployed using Docker and accessed via port 3000. The application was successfully launched and verified operational.

Access URL Format: http://<target-ip>:3000



6. Initial Reconnaissance Using Nmap

Initial reconnaissance was conducted using Nmap to identify open ports, running services, and operating system details of the target machine.

Commands Executed:

nmap -sS 192.168.230.131

nmap -sV 192.168.230.131

nmap -A -p- 192.168.230.131

```
┌──(root㉿kali)-[/home/divakar]
└─# nmap -sV -O -p- -sS 192.168.230.131
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-27 14:38 IST
Stats: 0:01:35 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 14:40 (0:00:03 remaining)
Nmap scan report for 192.168.230.131
Host is up (0.0013s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rexecd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
```

```
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
41619/tcp open  java-rmi    GNU Classpath grmiregistry
42642/tcp open  status      1 (RPC #100024)
51997/tcp open  nlockmgr    1-4 (RPC #100021)
54670/tcp open  mountd      1-3 (RPC #100005)
MAC Address: 00:0C:29:AD:2E:80 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.85 seconds
```

7. Key Findings from Initial Scan

Major Open Ports Identified:

• 21/tcp – FTP (vsftpd 2.3.4) – Critical Risk

• 22/tcp – SSH (OpenSSH 4.7p1) – Outdated Version

• 23/tcp – Telnet – Unencrypted Login

• 80/tcp – HTTP (Apache 2.2.8) – Outdated Server

• 445/tcp – SMB (Samba 3.x) – Remote Code Execution Risk

• 3306/tcp – MySQL – Weak Authentication Risk

8. Security Observations

The reconnaissance phase revealed multiple outdated services, default credentials, and unencrypted communication protocols. These vulnerabilities significantly increase the attack surface of the system.

If deployed in a production environment, these weaknesses could lead to remote code execution, unauthorized access, privilege escalation, and potential data breaches.

9. Conclusion

Week 1 successfully established a structured penetration testing lab and conducted comprehensive network reconnaissance. The exercise strengthened practical understanding of lab setup, service enumeration, attack surface identification, and professional documentation.